QUESTION 15: LATE LOAN PAYMENT PENALTY

AIM

To develop a COBOL program that reads an installment amount and the number of days late, calculates a penalty of 1% of the installment per late day (limited to a maximum of 10%), and displays the penalty and total payable to two decimal places.

ALGORITHM

1. Start the program.

2. Read the installment amount and number of days late.

3. Set the penalty rate to the number of days late, or 10 if the number of days exceeds 10. Thus, zero days late gives a zero penalty.

4. Calculate penalty = installment amount × penalty rate / 100.

5. Calculate total payable = installment amount + penalty.

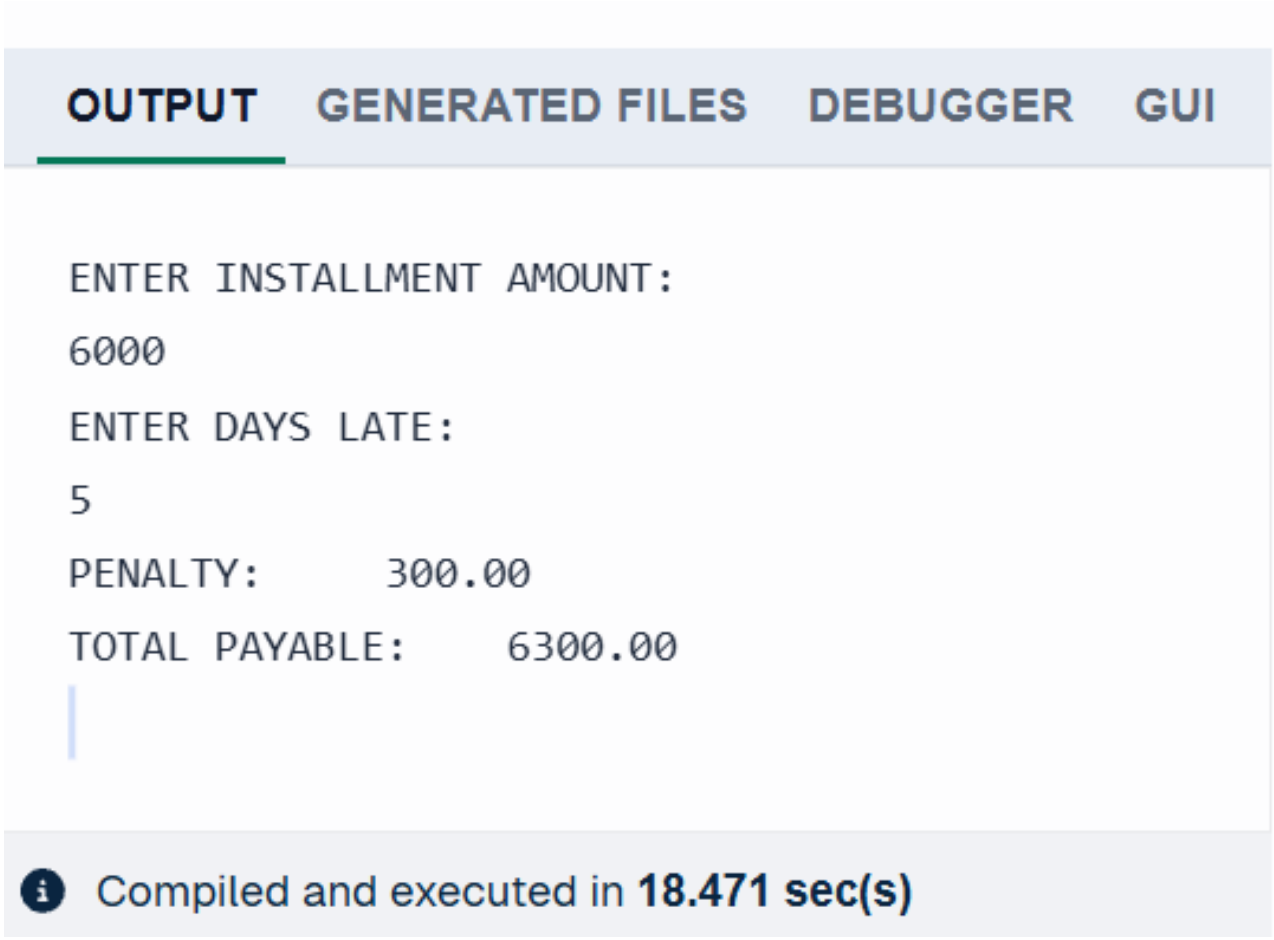
6. Display the penalty and total payable to two decimal places.

7. Stop the program.

PROGRAM

```
IDENTIFICATION DIVISION.
PROGRAM-ID. LOANPEN.
DATA DIVISION.
WORKING-STORAGE SECTION.
01 WS-AMOUNT PIC 9(7)V99.
01 WS-DAYS   PIC 999.
01 WS-RATE   PIC 99V99.
01 WS-FEE    PIC 9(7)V99.
01 WS-TOTAL  PIC 9(7)V99.
01 WS-FEE-O  PIC Z(6)9.99.
01 WS-TOT-O  PIC Z(6)9.99.
PROCEDURE DIVISION.
    DISPLAY "ENTER INSTALLMENT AMOUNT:"
    ACCEPT WS-AMOUNT
    DISPLAY "ENTER DAYS LATE:"
    ACCEPT WS-DAYS
    IF WS-DAYS > 10
        MOVE 10 TO WS-RATE
    ELSE
        MOVE WS-DAYS TO WS-RATE
    END-IF
    COMPUTE WS-FEE =
        WS-AMOUNT * WS-RATE / 100
    COMPUTE WS-TOTAL = WS-AMOUNT + WS-FEE
    MOVE WS-FEE TO WS-FEE-O
    MOVE WS-TOTAL TO WS-TOT-O
    DISPLAY "PENALTY: " WS-FEE-O
    DISPLAY "TOTAL PAYABLE: " WS-TOT-O
    STOP RUN.
```

OUTPUT



For the sample input of 6000 and 5 days late, the penalty is 300.00 and the total payable is 6300.00.